

Placement Related Questions

JAVA

(Total no. of Questions 224)

1. What is a class? What is an object? **[Nature Technologies]**
2. Explain the concept of OOPS? **[Nature Technologies]**
3. What are the advantages of OOPS over procedural programming?
4. What is Java?
5. What is the extension of a java file? Explain the features of Java.
6. Write a program to print a pyramid of star patterns. **[ICRA ANALYTICS]**
7. Explain the features of OOPS.. Discuss the five pillars of OOPS. **[Surelia]**
8. Write the comparison between Abstraction and Encapsulation.
9. Write the name of a pure object oriented programming language. Compare between Structured programming and object oriented programming.
10. Write a program in Java to find the Longest and second longest compound word from a text file. **[ICRA ANALYTICS]**
11. What is a top down approach? What is the bottom up approach?
12. **What is the bottom up approach?**
13. What are the main differences between the Java platform and other platforms?
14. What is byte code? What is JVM?
15. Write a program to count the total number of vowels and consonants in a string. **[ICRA ANALYTICS]**
16. What is a JIT compiler? State the steps to compile and run a java program .
17. Program to count the total number of vowels and consonants in a string **[ICRA ANALYTICS]**
18. What gives Java its 'write once and run anywhere' nature? Write a program to calculate the sum of diagonal of a matrix. **[ICRA ANALYTICS]**
19. State the differences between JDK, JRE, JVM. **[Web Idea Solution]**
20. Why is Java called platform independent?

21. Briefly explain different types of data types in Java?
22. What is the difference between a class and an object? **[Nature Technologies]**
23. If I don't provide any arguments on the command line, then what will the value stored in the String array passed into the main() method, empty or NULL?
24. What is inheritance?
25. What is encapsulation?
26. What is polymorphism? **[Sigmoid Technologies Solutions Private Limited]**
27. What is abstraction?
28. What are access modifiers? Explain the usage of them in JAVA.
29. What are the different types of variables in Java?
30. What are control structures in Java?
31. What is an enhanced for loop on Java?
32. What is a reference variable in JAVA?
33. What is function overloading?
34. What is function overriding?
35. How function overloading differs from function overriding?
36. Define Constructor? **[ICRA ANALYTICS]**
37. How many types of Constructors are in Java?
38. Write a Java Program to Copy the values from one object to another Object. Program to generate all factors of a number. **[ICRA ANALYTICS]**
39. Is there any method to call a subclass constructor from a superclass constructor?
40. Can we have a constructor in the Interface? Justify your answer.
41. Explain Constructor Chaining? (give example)
42. What is a private constructor?
43. Why constructors in Java cannot be static?
44. Can we make a constructor final?
45. Can we make a constructor abstract?
46. What is No-arg constructor? **[Deloitte]**
47. When do we need Constructor Overloading?
48. Do we have destructors in Java?
49. What will happen when a constructor is declared as protected?
50. Why is the constructor name similar to the class name?

51. Why is the return type not allowed for the constructor?
52. What is an array in Java? Write a program to sum even numbers from an array. **[Deloitte]**
53. What are the types of an array? Write a program to generate Fibonacci Series in Java **[TCS]**
54. Is it possible to declare array size as negative? Write Merge Sort Using Java **[TCS]**
55. What is the difference between `int array[]` and `int[] array`? Write a program in java to print Prime number between 120-180 **[TCS]**
56. How to copy an array in Java? find the unique no. of elements in an array and arrange the unique elements one after the other in the array **[Web Idea Solutions]**
57. What is the default value of the array? Can you store various data type in a single array in java ? **[Web Idea Solutions]**
58. What do you understand by the jagged array? What happens if we declare an array without assigning the size? Create a method will take integer as inputs – and the output will be of 2 - type string. example - int 2 string "1234567" output - array["12", "34", "56", "7"]
example - int 3 string "1234567" output - array["123", "456", "7"] **[ICRA ANALYTICS]**
59. Write a code (in any language) to implement stack data structure using array. **[Nature Technologies]**
60. Can we declare array size as negative? When `ArrayIndexOutOfBoundsException` occurs? Write a program to display "yes" if user is more than 18 years and display "no" if user is less than 18 year and negative number. **[Deloitte]**
61. What is the difference between Array and ArrayList? Write a program to concatenate str1 and str2 and print the resultant string. **[Deloitte]**
62. How can we check if an array contains values or not? In a given array of n size and print the sum of squares of the first two elements from the array. **[Deloitte]**
63. What are the advantages and disadvantages of an array? What are the differences between array and ArrayList. **[Surelia]**
64. What is the static variable? What is the static method? **[TCS]**
65. Write a program in Java to calculate the factorial of a given number. **[TCS]**
66. How can we run a java program without making any object? How to sort ArrayList. **[Surelia]**
67. Similarity and Difference between static block and static method ?
68. . How can we create objects if we make the constructor private ?
69. . Is it possible to compile and run a Java program without writing main() method?
70. . Can we initialize member variables within static block.

71. What is the purpose of 'this' keyword in java? Write a program in Java to implement Binary Search.**[Surelia]**
72. How do you implement inheritance practically in Java?
73. What is the purpose of inheritance?
74. What are the generalized and specialized classes in Java?
75. What are the types of inheritance?
76. What is the purpose of the 'super' keyword in Java?
77. What are the differences between 'this' and 'super' keyword?
78. What are the rules to be followed while overriding a method?
79. What is the base class of all classes? Explain multiple inheritance with an example.**[Nature Technologies]**
80. Does Java support multiple inheritances? Does it exists in Java **[Web Idea Solutions]**
81. How to define a constant variable in Java?
82. What is the purpose of declaring a variable as 'final'?What is the impact of declaring a method as final?
83. Explain call by value and call by reference with example.**[Nature Technologies]**
84. I don't want my class to be inherited by any other class. What should I do?
85. Can you give a few examples of final classes defined in Java API?
86. How is the final different from finally and finalize()? **[Surelia]**
87. Does a class inherit the constructors of its superclass?
88. What is Overriding?
89. How is this() and super() used with constructors?
90. What modifiers are allowed for methods in an Interface?
91. What are some alternatives to inheritance?
92. Does a class inherit the constructors of its superclass?
93. What restrictions are placed on method overloading?
94. What is method overloading & method overriding?
95. What is the difference between overloading & overriding?
96. What is the difference between the superclass and the subclass?
97. What modifiers may be used with top-level class?
98. What do you understand by an unreachable catch block error?
99. Can a class extend itself?

100. What is order of calling constructors in case of inheritance?
101. What happens if both superclass and subclass have a field with the same name?
102. Can we access both superclass and subclass members if we create an object of subclass?
103. Which of the following is correct way of inheriting class A by class B?
104. class B + class A { }
105. b) class B inherits class A { }
106. c) class B extends A { }
107. d) class B extends class A { }
108. Are static members inherited to sub classes?
109. What are abstract classes? What are the property of abstract calss ? Give an example of practical use of abstract class? **[Nature Technologies]**
110. Discuss Private Access vs Protected Access vs Public Access vs Default.
111. What is Object Wrapper and Autoboxing in java?
112. What is a Object Class in java?
113. What is the difference between compile-time polymorphism and runtime polymorphism?
114. What is Runtime Polymorphism?
115. Can you achieve Runtime Polymorphism by data members?
116. What is the difference between static binding and dynamic binding?
117. What is Java instanceof operator?
118. What is the difference between abstraction and encapsulation?
119. What is the abstract class?
120. Can there be an abstract method without an abstract class?
121. Can you use abstract and final both with a method?
122. Is it possible to instantiate the abstract class?
123. What is the interface?
124. Can you declare an interface method static?
125. Can the Interface be final?
126. What is a marker interface?
127. Can we define private and protected modifiers for the members in interfaces?
128. When can an object reference be cast to an interface reference?
129. How to make a read-only class in Java?
130. How to make a write-only class in Java?

131. What is the package?
132. What are the advantages of defining packages in Java?
133. How to create packages in Java?
134. How can we access some class in another class in Java?
135. Do I need to import java.lang package any time? Why?
136. Can I import same package/class twice? Will the JVM load the package twice at runtime?
137. What is the static import?
138. How many types of exception can occur in a Java program?
139. What is Exception Handling?
140. Explain the hierarchy of Java Exception classes?
141. What is the difference between Checked Exception and Unchecked Exception?
142. What is the base class for Error and Exception?
143. Is it necessary that each try block must be followed by a catch block?
144. What is finally block?
145. Can finally block be used without a catch?
146. Is there any case when finally will not be executed?
147. What is the difference between throw and throws?
148. Can an exception be rethrown?
149. Can subclass overriding method declare an exception if parent class method doesn't throw an exception?
150. What is exception propagation?
151. Can we have statements between try, catch and finally blocks?
152. What is Exception Chaining?
153. Can you catch and handle Multiple Exceptions in Java?
154. Differentiate between Checked Exception and Unchecked Exceptions in Java.
155. How do you handle checked exceptions?
156. What are runtime exceptions in Java?
157. What are the important methods defined in Java's Exception Class?
158. How are exceptions handled in Java?
159. What are some of the best practices to be followed while dealing with Java Exception Handling?
160. What are the rules we should follow when overriding a method throwing an Exception?

161. Are we allowed to use only try blocks without a catch and finally blocks?
162. Does the finally block always get executed in the Java program?
163. Under what circumstances should we subclass an Exception?
164. What are different scenarios where “Exception in thread main” types of error could occur?
165. What happens to the exception object after exception handling is complete?
166. Is it possible to throw checked exceptions from a static block?
167. What happens when an exception is thrown by the main method?
168. What does JVM do when an exception occurs in a program?
169. Explain Java Exception Hierarchy.
170. What is a nested class?
171. What are the advantages of Java inner classes?
172. What are the disadvantages of using inner classes?
173. What are the types of inner classes (non-static nested class) used in Java?
174. What are anonymous inner classes?
175. What is the nested interface?
176. Can a class have an interface?
177. Can an Interface have a class?
178. What is Garbage Collection?
179. What is gc()?
180. How is garbage collection controlled?
181. How can an object be unreferenced? What is the purpose of the Runtime class?
182. State the differences between Runtime and compile time Polymorphism. **[ICRA ANALYTICS]**
183. How will you invoke any external process in Java?
184. What are wrapper classes?
185. What is the thread?. **[ICRA ANALYTICS]**
186. Differentiate between process and thread?
187. What do you understand by inter-thread communication?
188. What is the purpose of wait() method in Java?
189. Why must wait() method be called from the synchronized block?
190. What are the advantages of multithreading?
191. What are the states in the lifecycle of a Thread?
192. What is the difference between preemptive scheduling and time slicing?

193. What is context switching?
194. Differentiate between the Thread class and Runnable interface for creating a Thread?
195. What does join() method?
196. Describe the purpose and working of sleep() method.
197. Discuss about the life cycle of a Thread.
198. What is the difference between wait() and sleep() method?
199. Is it possible to start a thread twice?
200. Can we call the run() method instead of start()?
201. What is shutdown hook?
202. When should we interrupt a thread?
203. What is the synchronization?
204. What is the purpose of the Synchronized block?
205. What is the difference between notify() and notifyAll()?
206. What is Thread Scheduler in java?
207. Does each thread have its stack in multithreaded programming?
208. What is race-condition?
209. What is the volatile keyword in java?
210. What do you understand by thread pool?
211. What is the difference between synchronous and asynchronous programming?
212. Between synchronised method and synchronised block, which do you prefer?
213. What is a thread scheduler, and how is it related to thread priority?
214. What is time slicing?
215. What is the busy spin technique?
216. What is thread starvation?
217. Why are the methods notify(), wait() and notifyAll() in object class?
218. How are the methods wait() and sleep() different?
219. Why is it important to override the run() method in a thread class?
220. Explain how you would stop the execution of a thread in Java.
221. What is the difference between sleep () and suspend()?
222. What are some fundamental advantages of multithreading in Java?
223. What are some functions used to perform inter-thread communication in Java?

DBMS

(Total no. of Questions 75)

1. What is Data?
2. State the differences between data and information.
3. What is Database?
4. What is DBMS?
5. State the differences between Database & Database Management System.
6. State the differences between Database & Data Structures.
7. What are the advantages of DBMS?

8. What are the advantages of DBMS over File System?
9. What are the disadvantages of DBMS?
10. What are the characteristics of DBMS?
11. Explain different languages present in DBMS.
12. What is CRUD in database?
13. Define 3 schema architecture in DBMS?
14. What is Data Independence?
15. How many levels of abstraction can be achieved in DBMS?
16. What is Instance and schema?
17. What is the role of Data Dictionary?
18. What are the languages used in DBMS?
19. How many types of users are there in DBMS? What is the difference between DDL and DML.**[Surelia]**
20. What are the five major functions of a database administrator?
21. Compare relational, network and hierarchical data models by stating the advantages and disadvantages of every model compared to each other.
22. What is an Entity?
23. What integrity rules exist in a DBMS?
24. What are the components of E-R Model?
25. Briefly discuss the components of E-R Model?
26. Briefly discuss the following terms used in DBMS Attribute, Tables, Tuple Relation Schema, Degree, Cardinality, Column Relation instance, Relation key, Attribute domain.**[TCS]**
27. What is the role of ER Model to create a database?
28. Give an example of each of the following relationships:
 - a. one to one
 - b. one to many
 - c. many to many
29. What are the differences between generalization and specialization?
30. What is Aggregation? State with an example.
31. What are the various types of attributes? Discuss about them.
32. What are CODD's Rule? Briefly discuss about each rule.
33. How many rules should a software satisfy to qualify as a RDBMS?

34. What is Integrity rule? Discuss briefly about them?
35. What is Relational Algebra?
36. What is query language? How many types of query languages are present?
37. What is the role of relational algebra in query language?
38. Briefly discuss about various set operations and relational algebra operations?
39. What is a JOIN operation? / Explain joining . **[Web Idea Solutions, Nature TEchnologies]**
40. Give simple example to example various types of join present in relational algebra? Inner Join & Outer Join **[TCS]**
41. What are the differences between Tuple Relation Calculus and Domain Relation Calculus?
42. What is a key and discuss the role of various types of keys in DBMS? How to add foreign key **[Sigment Technologies Solutions Private Limited]**
43. What is Lossless and Lossy Join Decomposition?
44. What is a Normalization? **[Web Idea Solutions]**
45. Sate the need of Normalization? Difference between left join . **[Surelia]**
46. How normalization differ from denormalization?
47. What are the different types of anomalies?
48. What are the various types of normalization levels or normalization forms?
49. What is the First Normal Form (1NF) in database normalization?
50. What is an atomic value, and how does it relate to 1NF?
51. What is the Second Normal Form (2NF), and why is it an improvement over 1NF? How can you normalize a table from 1nf to 2nf? **[Nature Technologies]**
52. Explain partial dependency and provide an example.
53. What is the Third Normal Form (3NF), and how does it further refine the database schema?
54. Explain transitive dependency and provide an example.
55. What is the Boyce-Codd Normal Form (BCNF), and when is it applied in database design?
56. E What are prime and non-prime attributes in the context of BCNF?xplain the concept of a candidate key in the context of BCNF.
57. What is denormalization, and when might it be used in database design?
58. What is the purpose of the Fourth Normal Form (4NF), and when is it applied in database design?
59. What are multi-valued dependencies, and how do they relate to 4NF?

60. What is the Fifth Normal Form (5NF), and when is it used in database design?
61. What is the redundancy involved in a non-normalized (unnormalized) database, and why is it a concern?
62. How does normalization help prevent data anomalies in a database?
63. What are the potential downsides of over-normalization in database design?
64. What is the difference between functional dependency and full functional dependency?
65. What are the primary and secondary objectives of normalization in database design?
66. Explain the process of identifying functional dependencies in a database table.
67. What is a non-prime attribute, and how does it relate to normalization?
68. What is the role of dependency preservation in normalization, and how is it achieved?
69. What is the role of denormalization, and when might it be used despite the benefits of normalization?
70. How do you determine the appropriate level of normalization for a specific database design?
71. What is the purpose of a data dictionary in database design, and how does it relate to normalization?
72. Explain the concept of closure in functional dependencies and how it is used in normalization.
73. What is the difference between functional dependency and multivalued dependency in database normalization?
74. What is Armstrong Axiom? What is ACID property. **[Sigmoid Technologies Solutions Private Limited]**
75. What is transitive dependency?

SQL

(Total no. of Questions 75)

1. What is SQL?
2. What are SQL dialects? Give some examples.
3. What are the main applications of SQL?

4. What is an SQL statement? Give some examples.
5. What types of SQL commands (or SQL subsets) do you know?
6. Give some examples of common SQL commands of each type.
7. What are tables and fields in SQL?
8. What is an SQL query, and what types of queries do you know?
9. What is a subquery?
10. What types of SQL subqueries do you know?
11. What is a constraint, and why use constraints?
12. What SQL constraints do you know?
13. What is a SQL comment?
14. What types of SQL operators ?
15. What is an alias?
16. What is a clause?
17. What are some common statements used with the SELECT query?
18. How to delete a table from a database?
19. How to get the count of records in a table?
20. How to sort records in a table?
21. How to select all columns from a table?
22. How to select common records from two tables?
23. What is the DISTINCT statement and how do you use it?
24. What is NULL value? How is it different from zero or a blank space?
25. What are the various types of SQL functions ?
26. What are aggregate functions in SQL? Give some examples.
27. What are scalar functions in SQL? Give some examples.
28. What are case manipulation functions? Give some examples.
29. What are character manipulation functions? Give some examples.
30. What is the default data ordering with the ORDER BY statement, and how do you change it?
31. What set operators are used in SQL?
32. What operator is used in the query for pattern matching?
33. What is the order of appearance of the common statements in the SELECT query?
34. What is a view, and why use it?
35. Can we create a view based on another view? Give Example

36. Can we still use a view if the original table is deleted?
37. What is the difference between renaming a column and giving an alias to it?
38. What is the difference between nested and correlated subqueries?
39. What is the CASE() function?
40. What is the difference between the DELETE and TRUNCATE statements?
41. What is the difference between the DROP and TRUNCATE statements?
42. What is the difference between the HAVING and WHERE statements?
43. How do you add a record to a table?
44. How to find the nth highest value in a column of a table?
45. How to find the values in a text column of a table that start with a certain letter?
46. How to find the last id in a table?
47. How to select random rows from a table?
48. How to prevent duplicate records when making a query? Write a SQL query for update the name in the table[arhat Iserv LLP (Responses)]
49. How to select all even or all odd records in a table?
50. What do you mean by a NULL value in SQL?How to join two tables[ICRA ANALYTICS]
51. How to find the second highest salary from a table in SQL?Define a Unique Key in SQL [ARC Documents]
52. Write query to find the employee name who are getting salary grater than department wise avg salary .[Nature Technologies]
53. Explain the differences between SQL and NoSQL databases.
54. What is a constraint in SQL? Name a few
55. What are indexes in SQL?Explain GROUP BY in SQL
- 56..Write query for finding customer id who order maximum time from order table .[Nature Technologies]
57. What is an SQL alias?
58. What are aggregate functions? Can you name a few?
59. How do you update a value in SQL?
60. What is a subquery? Provide an example.
61. How do you optimize SQL queries?
62. What is the difference between UNION and UNION ALL?
63. What is the difference between CHAR and VARCHAR2 datatypes in SQL?

- 64. What is a Default constraint?
- 65. What is Auto Increment?
- 66. What are Union, minus, and Intersect commands?
- 67. What is a T-SQL?
- 68. How to copy tables in SQL?
- 69. What are the differences between SQL and PL/SQL and MySQL?
- 70. What is the difference between BETWEEN and IN operators in SQL?
- 71. Write an SQL query to find the names of employees starting with 'A'.
- 72. What is the On Delete cascade constraint?
- 73. Explain WITH clause in SQL?
- 74. Define BETWEEN statements in SQL?
- 75. Why do we use Commit and Rollback commands?

[N.B. For more commands in SQL students are advised to explore various case studies and practice.]

C

(Total no. of Questions 113)

1. What are the features of the C programming language?
2. Difference between Compiler and interpreter. [Nature Technologies]
3. What is the meaning of #include in a C program?
4. What is the difference between #include <filename> and #include "filename"? [CTS]
5. What are header files and their uses?
6. Why is C called a mid-level programming language?
7. What is the use of printf() and scanf() functions in C Programming language? Also, explain format specifiers.
8. What is the use of a semicolon (;) at the end of every program statement?
9. What is volatile in C?
10. What is typecasting in C?
11. What are bitwise operators? Provide examples.
12. What happens if you don't use a break statement in a switch case?
13. Why do we use default statements in switch cases in C?
14. Explain the if, if-else, and else-if statements in C.
15. How do loops work in C? Explain for, while, and do-while loops.
16. How can you exit from a loop in C?
17. What are macros in C?

18. What are command line arguments?
19. What is the difference between macro and functions?
20. What is the difference between call by value and call by reference? **[TCS,CTS,IBM]**
21. What is pass-by-reference in functions?
22. What are goto statements in C? Should we use them?
23. How to return array in C?
24. What is an array in C?
25. How do you declare and initialize an array in C?
26. Explain the syntax for accessing array elements.
27. What is the difference between a one-dimensional array and a multi-dimensional array?
28. How do you determine the size of an array?
29. Write a Program to sort a string of characters [Infosys]
30. Write a program to count the number of unique characters in a given string. **[Infosys]**
31. How do you initialize a one-dimensional array with values at the time of declaration?
32. How do you iterate over the elements of a one-dimensional array?
33. Write a C program to find the maximum and minimum element in a one-dimensional array.
34. How do you pass a one-dimensional array to a function?
35. How do you return an array from a function?
36. How do you declare and initialize a two-dimensional array in C?
37. How do you access elements of a two-dimensional array?
38. How do you pass a two-dimensional array to a function? **[TCS]**

39. How do you work with multi-dimensional arrays beyond two dimensions?
40. How do you declare and initialize a string in C?
41. What is the difference between character arrays and strings in C? **[CTS]**
42. How do you read and print strings in C?
43. How do you concatenate two strings in C?
44. Write a C program to find the length of a string without using the strlen function.
45. What is array of pointer and pointer of array
46. How would you find the smallest and largest element in an array?
47. What are tokens in C?
48. What is the difference between continue and break statements in C?
49. What is the difference between pre-increment and post-increment in C? **[TCS]**
50. What is the difference between type casting and type conversion?
51. What are basic data types supported in the C Programming Language?
52. How to convert a string to numbers in C? **[CTS]**
53. What is the difference between the local and global variables in C?
54. What is typedef in C?
55. What are enumerations?
56. how can we create an infinite loop in C?
57. What is the difference between getc(), getchar(), getch() and getche(). **[TCS,CTS]**
58. What is recursion in C?
59. What are preprocessor directives in C? **[TCS,IBM]**

60. Can we compile a program without a main() function? **[CTS]**
61. What is the use of static variables in C?
62. What is the difference between malloc() and calloc() in the C programming language?
63. What do you mean by dangling pointers and how are dangling pointers different from memory leaks in C programming? **[IBM]**
64. What is the Pointer to Pointer in C?
65. What is a pointer and how do you declare it?
66. Explain pointer arithmetic.
67. What are null pointers and void pointers?
68. How do you pass pointers to functions?
69. What is a NULL pointer in C? **[TCS,CTS,IBM]**
70. What are huge pointers?
71. What is dynamic data structure?
72. What is near, far, and huge pointers in C? **[TCS,IBM]**
73. What is a memory leak and how to avoid it? **[IBM]**
74. What is a structure and Union?
75. What is the difference between struct and union in C?
76. What is an r-value and l-value? **[CTS]**
77. What are the basic data types in C?
78. What is a pointer?
79. What is the difference between reference and pointer? **[Infosys]**

80. Explain the difference between malloc and calloc.
81. What is a segmentation fault?
82. What is the purpose of the sizeof operator?
83. How do you declare a constant in C?
84. What are the storage classes in C?
85. What is the difference between == and =? **[TCS]**
86. What is a null pointer?
87. What is array of pointer and pointer of array **[Nature Technologies]**
88. How do you pass a pointer to a function?
89. What is a function pointer? **[Infosys]**
90. What is the purpose of the volatile keyword?
91. How do you dynamically allocate memory in C?
92. What is the purpose of the static keyword? **[CTS]**
93. What is recursion? Give an example.
94. What is the difference between stack and heap memory? **[TCS]**
95. What is a memory leak? How can you avoid it?
96. Explain the difference between deep copy and shallow copy. **[TCS,CTS,IBM]**
97. What is the purpose of the extern keyword?
98. How does the const keyword differ when applied to pointers?
99. What is the difference between fgets and gets?
100. How do you handle errors in C?

101. How do you declare a function prototype?
102. What is a union in C, and when would you use it?
103. What is the purpose of the goto statement?
104. How to convert string to integer value. **[Deloitte]**
105. How do you handle file operations in C?
106. What are the differences between printf and sprintf?
107. What is the static function scope?
108. How do you open, read, write, and close files in C?
109. What is the difference between text mode and binary mode file operations?
110. How do you handle errors during file operations?
111. What is array of structure.
112. What is pointer array of structures in c **[Infosys]**
113. Write compilation process in c **[TCS,CTS]**

Data Structures

(Total no. of Questions 116)

1. Can an array be resized at runtime?

2. What is Data Structures? **[CTS]**
3. What is ADT **[TCS,CTS,Infosys]**
4. What are some applications of Data structures? **[TCS]**
5. Describe the types of Data Structures?
6. What is a Linear Data Structure? Name a few examples.
7. What is asymptotic notation in data structure **[TCS,CTS,IBM,Infosys]**
8. What is time complexity?
9. What is space complexity?
10. What is the time complexity for accessing an element in an array?
11. What is the difference between an array and a linked list? **[TCS]**
12. Discuss the advantages and disadvantages of using arrays.
13. What is a linked list?
14. What are the different types of linked lists?
15. What are the advantage of Linked List?
16. What are the disadvantages of Linked List?
17. How do you reverse a linked list?
18. How do you detect a cycle in a linked list?
19. How do you find the middle element of a linked list?
20. How do you merge two sorted linked lists?
21. How do you remove duplicates from an unsorted linked list?

22. What is the time complexity of common linked list operations (insertion, deletion, and search)?
23. Compare the time and space complexities of singly linked lists and doubly linked lists.
24. What are the pros and cons of using a circular linked list?
25. What are different operations available in stack data structure?
26. What is a queue data structure? What are the applications of queue? **Accenture**
27. How would you compare dynamic arrays Vs linked Lists?
28. What is a stack?
29. What are the operations performed on a stack?
30. How is a stack implemented in an array?
31. What is the time complexity of stack operations?
32. What are the applications of a stack?
33. What is a stack overflow? **Accenture**
34. What is a stack underflow?
35. What is a postfix expression?
36. How can a stack be used to evaluate a postfix expression?
37. What is a prefix expression?
38. How can a stack be used to evaluate a prefix expression?
39. What is a Queue?
40. What are the different types of Queues?
41. How is a Queue implemented in an array?

42. How is a Queue implemented in a linked list?
43. What is the time complexity of enqueue and dequeue operations in a Queue?
44. What is the difference between a Queue and a Stack?
45. What are the applications of Queues?
46. How do you handle overflow and underflow conditions in a Queue?
47. What is a circular and priority queue? **[TCS]**
48. What is a double-ended queue (Deque)?
49. How is a deque implemented?
50. How do you delete a node from a singly linked list?
51. Implement a function to split a linked list into two halves.
52. How would you convert a binary tree to a doubly linked list?
53. Write a function to add two numbers represented by linked lists. **[TCS]**
54. How can you optimize a sorting algorithm for nearly sorted data?
55. How do you implement a stack using an array?
56. How do you implement a queue using a linked list?
57. How do you detect a cycle in a linked list?
58. How do you insert a node at the beginning and end of a singly linked list?
59. What is a heap data structure?
60. What are the two types of heaps?
61. What is a tree data structure?
62. What is a B-tree, and how is it different from a binary search tree? **[TCS]**

63. Explain how a B+ tree works and its advantages over a B-tree.
64. Compare the time and space complexities of sorting using a BST, AVL Tree
65. What is Inorder, Preorder and postorder traversal. [TCS]
66. What is the impact of tree height on the performance of sorting algorithms using trees?
67. How does balancing affect the efficiency of sorting in tree-based structures?
68. Implement an AVL tree and explain how it maintains balance during insertions and deletions.
69. What is the time complexity of inserting an element into a heap?
70. List the types of trees?
71. Write a algo sum of right node of a Binary tree [Chikpea]
72. Implementation of binary tree [Chikpea]
73. Calculate sum all right leaves of a BST. [Chikpea]
74. What is binary tree data structure?
75. What are the applications for binary search trees?
76. What are tree traversals?
77. What is the difference between the Breadth First Search (BFS) and Depth First Search (DFS)?
78. What is a Tree?
79. Explain different types of trees.
- 80. What are the basic operations performed on a tree? [TCS]**
81. What are the different ways to represent a tree in memory?
82. What are the advantages and disadvantages of using trees?
83. What are Binary trees?

84. How to add element in Binary search tree. **[TCS,CTS,Infosys]**
85. How to delete a element in Binary search tree **[TCS,CTS,Infosys]**
86. Difference between Binary tree and Binary search tree? **[IBM]**
87. Construct the given binary search tree and get the sum of all leaf nodes **[Chikpea]**
88. write a code of binary tree to add the node of left of the node of the tree **[Chikpeab]**
89. What is B and B+ tree? **[TCS]**
90. What are the differences between the B tree and the B+ tree?
91. What are the advantages of binary search over a linear search? **[IBM]**
92. Explain the concept of a binary search tree.
93. How can you convert a binary search tree into a sorted array?
94. What is AVL tree data structure, its operations, and its rotations? **[TCS]**
95. What are the applications for AVL trees?
96. Which sorting algorithm is considered the fastest? Why? **[TCS,CTS,Infosys,IBM]**
97. How does the Selection sort work?
98. What are some examples of divide and conquer algorithms? **[TCS,CTS]**
99. What is sorting?
100. What is the difference between comparison-based and non-comparison-based sorting algorithms?
101. Explain how Bubble Sort works. **[ICRA ANALYTICS]**
102. Write algo merge sort. **[TCS, Accenture]**
103. What is the time complexity of Quick Sort in the best, average, and worst case?

- 104. How does Merge Sort work, and what is its time complexity?
- 105. What is the main advantage of Heap Sort over other sorting algorithms?
- 106. Compare the time and space complexities of different sorting algorithms
- 107. How does the choice of pivot affect the performance of Quick Sort?
- 108. Explain how to sort a linked list.
- 109. What is the impact of sorting on cache performance and memory usage?
- 110. Explain how a binary tree is constructed. **[TCS, Blue Flame Labs]**
- 111. Let a complete binary tree has root node. Let if a node is s then its left child will $2s$ and right child is $2s+1$. Then make a function to find shortest path distance between two nodes.
[Blue Flame Labs]
- 112. Write algo of linear search.
- 113. What is a binary search algorithm? **[TCS,CTS]**
- 114. How do you perform a depth-first search on a graph?
- 115. How do you find the length of a string without using the strlen function?
- 116. Write algo of quicksort? **[TCS,IBM]**

PHP & MySQL

(Total no. of Questions 164)

1. What is PHP? [Google - 2022]
2. Explain the differences between PHP 5 and PHP 7. [**Facebook - 2022.**]
3. What are the common uses of PHP? [**LinkedIn - 2016.**]
4. How do you declare a variable in PHP? [Google - 2023]
5. What is the scope of a variable in PHP? [Microsoft - 2021]
6. How do you write comments in PHP? [**LinkedIn - 2016.**]
7. What is the difference between single-quoted and double-quoted strings in PHP? [**Amazon - 2020.**]
8. Explain the use of echo and print statements. [Facebook - 2022]
9. What are PHP constants and how do you define them? [**IBM - 2019**]
10. How do you include files in PHP? [Microsoft 2014.]
11. What are the different data types supported by PHP?[TechInnovate 2022]
12. How do you check the type of a variable in PHP?[Google 2020.]
13. Explain type juggling in PHP.[Microsoft 2023]
14. Explain the different types of loops in PHP.
15. What is the difference between while and do-while loops?[Google 2019.]
16. How do you use foreach loop in PHP?
17. Explain the use of switch statement.[IBM, Google, and TCS]
18. What is the difference between break and continue statements?[]
19. How do you define and use arrays in PHP?[TechInnovate 2024.]
20. What is the difference between indexed and associative arrays?[]
21. How do you define a function in PHP?[**TechInnovate 2024.**]
22. What is the purpose of return statement in a function?
23. Explain variable scope within a function.
24. How do you pass arguments to a function?
25. What are default arguments in a function?
26. What is object-oriented programming in PHP?
27. Explain the concept of classes and objects.
28. How do you define and use constructors and destructors?
29. What is inheritance and how is it implemented in PHP?

30. Explain the concept of interfaces in PHP.
31. What are the different error types in PHP?
32. How do you handle errors in PHP using try-catch blocks?[TechInnovate 2024]
33. Explain the use of `set_error_handler()` function.
34. What is the purpose of `trigger_error()` function? [Tech Innovate company interview in 2024.]
35. How do you handle fatal errors in PHP?
36. What are sessions in PHP and how do you start a session?[IBM 2019]
37. How do you store and retrieve session variables?
38. What are cookies and how do you set them in PHP?
39. How do you retrieve and delete cookies in PHP?
40. What are the differences between sessions and cookies?
41. How do you open and close files in PHP?
42. What are the different modes of opening a file?
43. How do you read and write files in PHP?
44. Explain the use of `fopen()`, `fread()`, and `fwrite()` functions.
45. How do you check if a file exists in PHP?
46. What are some common string functions in PHP?
47. How do you concatenate strings in PHP?
48. Explain the use of `strlen()` and `strpos()` functions.
49. How do you replace part of a string using `str_replace()`?
50. What is the purpose of `substr()` function?
51. What are some common array functions in PHP?
52. How do you merge two arrays in PHP?
53. Explain the use of `array_push()` and `array_pop()`.
54. How do you find the length of an array?
55. What is the purpose of `array_search()` function?
56. How do you get the current date and time in PHP?
57. Explain the use of `date()` function.
58. How do you format a date in PHP?
59. What is the purpose of `strtotime()` function?
60. How do you calculate the difference between two dates?
61. What is the purpose of the `$_GET` and `$_POST` arrays?

62. How do you send an email using PHP?
63. Explain the use of header() function.
64. How do you handle file uploads in PHP?
65. What is the purpose of \$_SESSION and \$_COOKIE arrays?
66. What are some common security threats in PHP?
67. How do you prevent SQL injection in PHP?
68. What is MySQL?
69. Explain the differences between MySQL and SQL.
70. What is a database?
71. What are tables in a database?
72. What is a primary key?
73. What are the different data types supported by MySQL?
74. Explain the difference between CHAR and VARCHAR.
75. What is the TEXT data type used for?
76. What are the different numeric data types in MySQL?
77. What is the DATE data type used for?
78. How do you create a database in MySQL?
79. How do you create a table in MySQL?
80. Explain the use of SELECT statement.
81. How do you insert data into a table?
82. What is the purpose of UPDATE statement?
83. What are joins in SQL?
84. Explain the different types of joins in MySQL.
85. What is the difference between INNER JOIN and OUTER JOIN?
86. How do you perform a LEFT JOIN?
87. What is a CROSS JOIN?
88. What is an index in MySQL?
89. What are the different types of indexes?
90. How do you create an index on a table?
91. What is the purpose of UNIQUE index?
92. How do indexes improve query performance?
93. What are constraints in SQL?

94. What is a PRIMARY KEY constraint?
95. Explain the use of FOREIGN KEY constraint.
96. What is a UNIQUE constraint?
97. How do you define a CHECK constraint?
98. What are stored procedures?
99. How do you create a stored procedure in MySQL?
100. What is a trigger in MySQL?
101. How do you create a trigger?
102. What are views in MySQL?
103. What is a transaction in SQL?
104. Explain the use of COMMIT and ROLLBACK.
105. What are the ACID properties?
106. How do you start a transaction in MySQL?
107. What is the purpose of SAVEPOINT?
108. How do you optimize SQL queries?
109. What is query caching in MySQL?
110. Explain the use of EXPLAIN statement.
111. What are the best practices for indexing?
112. How do you optimize database schema?
113. How do you connect to a MySQL database using PHP?
114. What is mysqli and how is it different from mysql?
115. How do you execute a SQL query in PHP?
116. Explain the use of PDO in PHP.
117. How do you fetch data from a MySQL database in PHP?
118. How do you handle MySQL connection errors in PHP?
119. What is the purpose of mysqli_error() function?
120. How do you handle SQL query errors in PHP?
121. What are the common MySQL error codes?
122. How do you log MySQL errors in PHP?
123. How do you prevent SQL injection attacks in PHP?
124. What are prepared statements and how do you use them in PHP?
125. How do you secure MySQL user accounts?

126. What is the purpose of `mysqli_real_escape_string()`?
127. How do you handle sensitive data in MySQL?
128. What is the purpose of LIMIT clause in SQL?
129. How do you sort query results in MySQL?
130. What is the use of GROUP BY clause?
131. How do you use aggregate functions in MySQL?
132. What is the purpose of HAVING clause?
133. How do you insert data into a MySQL database using PHP?
134. How do you update records in a MySQL table using PHP?
135. How do you delete records from a MySQL table using PHP?
136. How do you handle multiple queries in PHP?
137. What are the common MySQL functions in PHP?
138. What is ORM (Object-Relational Mapping)?
139. How do you use an ORM in PHP?
140. What are the benefits of using an ORM?
141. What is database normalization?
142. Explain the different normal forms.
143. How do you analyze slow queries in MySQL?
144. What are the best practices for writing efficient SQL queries?
145. How do you use MySQL profiling?
146. What is the purpose of EXPLAIN ANALYZE?
147. How do you use indexing to improve performance?
148. What are PHP superglobals?
149. What is the purpose of the `$_GET` superglobal?
150. What is the difference between include and require in PHP?
151. How do you connect to a MySQL database in PHP?
152. What is the `isset()` function used for in PHP?
153. How do you handle errors in PHP?
154. What is a PHP namespace?
155. What is the difference between INNER JOIN and LEFT JOIN?
156. What is normalization in MySQL?
157. What are MySQL transactions?

- 158. Write a program in PHP to INSERT data in a database file.
- 159. Write a program in PHP to DISPLAY data from a database file.
- 160. Write a program in PHP to DELETE data from a database file.
- 161. Write a program in PHP to UPDATE data from a database file.
- 162. Write a program in PHP to prepare a Registration form using validation.
- 163. Write a program in PHP to prepare a Login form using validation.
- 164. How do you handle user authentication in PHP?

Java Script

(Total no. of Questions 139)

1. What is JavaScript?
2. What are the different data types in JavaScript?
3. Explain the difference between var, let, and const.
4. What are the advantages and disadvantages of using JavaScript?
5. What is hoisting in JavaScript?
6. What is the this keyword in JavaScript?
7. What are JavaScript data types?
8. Explain == vs ===.
9. What is a closure?
10. Explain the difference between function declarations and function expressions.
11. What are arrow functions?
12. What is the this keyword in JavaScript?
13. Explain the concept of scope in JavaScript.
14. What are the different ways to iterate over an array in JavaScript?
15. Explain the concept of prototypal inheritance.
16. What is the difference between null and undefined?
17. How do you create an object in JavaScript?
18. How do you clone an object in JavaScript?
19. What is an event loop in JavaScript?
20. Explain the difference between synchronous and asynchronous code.
21. What are Promises in JavaScript?
22. How do you handle errors in Promises?
23. What is async/await in JavaScript?
24. What is the DOM?
25. How do you select an element by ID in JavaScript?
26. How do you select elements by class name in JavaScript?
27. How do you select elements using a CSS selector in JavaScript?

28. How do you create an element in JavaScript?
29. How do you add an element to the DOM?
30. How do you remove an element from the DOM?
31. How do you change the content of an HTML element?
32. How do you change the style of an HTML element?
33. What is event delegation?
34. What is a higher-order function in JavaScript?
35. Explain the concept of callback functions.
36. What is the bind method in JavaScript?
37. What is the difference between call and apply?
38. What is memoization?
39. What is the event loop and how does it work?
40. What are modules in JavaScript?
41. What is the difference between ES5 and ES6?
42. What are promises and why are they used?
43. What is async/await and how does it work?
44. What are design patterns and why are they important?
45. What is the Module pattern?
46. What is the Observer pattern?
47. What is the Factory pattern?
48. What is the difference between deep copy and shallow copy?
49. What are web workers and why are they used?
50. What is CORS and how does it work?
51. What is the difference between synchronous and asynchronous code execution?.
52. How can you create an immediately invoked function expression (IIFE)?
53. Explain event delegation in JavaScript.
54. What is the difference between null and undefined?
55. What are arrow functions, and how do they differ from regular functions?
56. How do you create a promise in JavaScript?
57. What is the purpose of the async and await keywords?
58. What is a higher-order function?
59. Explain the concept of hoisting in JavaScript.

60. What is the difference between slice() and splice() methods in JavaScript?
61. How can you prevent an object from being modified?
62. What is the prototype property in JavaScript?
63. How does the bind() method work?
64. What is the difference between map() and forEach()?
65. Explain the concept of promises chaining.
66. How do you handle errors in promises?
67. What is the purpose of the typeof operator?
68. How can you check if a variable is an array in JavaScript?
69. What is the arguments object in JavaScript?
70. Explain the concept of function currying.
71. What is the difference between synchronous and asynchronous code?
72. How do you handle asynchronous code in JavaScript?
73. What is a Symbol in JavaScript?
74. Explain the concept of debouncing in JavaScript.
75. What is the event loop in JavaScript?
76. How does JavaScript handle asynchronous operations?
77. What is the spread operator in JavaScript?
78. Explain the concept of rest parameters.
79. How do you create a deep copy of an object?
80. What is the Proxy object in JavaScript?
81. Explain the difference between call() and apply().
82. What is event bubbling and event capturing?
83. How can you stop event propagation?
84. What is the purpose of the defaultPrevented property?
85. How do you create a custom event in JavaScript?
86. What are service workers?
87. How do you detect if cookies are enabled in the browser?
88. Explain the fetch API.
89. What is the difference between localStorage and sessionStorage?
90. How do you set a timeout in JavaScript?
91. What is the reduce() method in JavaScript?

- 92. How do you check if a variable is a function?
- 93. What is the instanceof operator?
- 94. How do you compare two objects in JavaScript?
- 95. What is a WeakMap?
- 96. How can you convert an array-like object to an array? ARRAY LIST TO FIND THE MAX NUMBER
[Web Idea Solutions]
- 97. Explain the Promise.all() method.
- 98. How can you debounce a function in JavaScript?
- 99. What is the difference between Array.prototype.forEach and How can you throttle a function in JavaScript? find the unique no. of elements in an array and arrange the unique elements one after the other in the array **[Web Idea Solution]**
- 100. What is the Reflect object in JavaScript?
- 101. How do you use the spread operator to copy an array? ARRAY LIST TO FIND THE MAX NUMBER. **[Web Idea Solution]**
- 102. What is a tagged template literal?
- 103. Explain the concept of the temporal dead zone.
- 104. What is the difference between Object.seal() and Object.freeze()?
- 105. How can you shallow copy an object?
- 106. What is the purpose of the Promise.race() method?
- 107. How do you convert a string to a number in JavaScript?
- 108. What is the difference between Function.prototype.call and Function.prototype.apply?
- 109. How do you remove duplicates from an array? You are given two integer arrays nums1 and nums2, sorted in non-decreasing order, and two integers m and n, representing the number of elements in nums1 and nums2 respectively. Merge nums1 and nums2 into a single array sorted in non-decreasing order. **[Web Idea Solutions]**
- 110. Explain the difference between window.onload and document.ready.
- 111. What are JavaScript generators?
- 112. How do you handle deep comparisons in JavaScript?
- 113. What is a template literal?
- 114. How do you create a multi-line string in JavaScript?
- 115. What is an ES6 module?
- 116. What is a WeakSet?

117. Explain the concept of memoization.
118. How do you detect the user's browser in JavaScript?
119. What is the purpose of the `requestAnimationFrame` method?
120. How do you implement inheritance in JavaScript?
121. What is a Proxy in JavaScript?
122. How do you debounce a function in JavaScript?
123. Explain the concept of currying.
124. What is the difference between `Function.prototype.bind` and `Function.prototype.call`?
125. How can you create a custom event in JavaScript?
126. What is the difference between undefined and not defined?
127. How do you create a private variable in JavaScript?
128. What is the purpose of the `Object.create` method?
129. Write a JavaScript function to add two numbers and return the result.
130. Write a JavaScript function that checks if a number is even or odd.
131. Write a JavaScript function to print numbers from 1 to 10 using a for loop.
132. Write a JavaScript function to find the largest number in an array.
133. Write JavaScript code to validate a form input for a non-empty value before submitting.
134. Write JavaScript code to handle errors using try/catch.

HTML

(Total no. of Questions 104)

1. Where should the `<!DOCTYPE html>` declaration be placed in an HTML document?
2. How do you specify the character set for an HTML document?
3. What is an HTML element?
4. Where should attributes be placed within an HTML tag?
5. How do you add a class to an HTML element?
6. How do you add an ID to an HTML element?
7. Name three elements that are typically placed within the `<head>` section.
8. Can you include multiple `<body>` tags in a single HTML document?
9. How do you emphasize text in HTML?
10. What is the difference between `<b>` and `<strong>` tags?
11. What is the purpose of the `<section>` element?
12. When should you use the `<article>` element?
13. Can you nest lists within other lists in HTML?
14. What attribute is used to specify the destination of a link?
15. How do you open a link in a new tab?
16. What is the purpose of the `<nav>` element?
17. How do you embed an image in an HTML document?
18. How do you embed a video in an HTML document?
19. What element is used to embed audio in an HTML document?
20. What attribute specifies where to send the form data?
21. What element is used to group related form controls?

22. How do you merge two or more cells in a table?
23. How do you specify metadata for an HTML document?
24. What is the purpose of the <meta> tag?
25. How do you set the viewport for responsive web design?
26. How do you link an external CSS file to an HTML document?
27. What is the <canvas> element used for?
28. How do you include a script in an HTML document?
29. What is the purpose of the <noscript> element?
30. How do you create a dropdown list in a form?
31. What is the <template> element used for?
32. Why is it important to use semantic HTML tags?
33. How can you improve the accessibility of an HTML document?
34. What are some best practices for optimizing HTML for performance?
35. How do you handle browser compatibility issues in HTML?
36. What tools and validators can you use to ensure your HTML code is error-free and follows standards?
37. How do you create a hyperlink that opens in a new tab?
38. What is the purpose of the target attribute in the <a> tag?
39. How do you create an anchor link to navigate within the same page?
40. What is the <nav> tag used for?
41. How do you create a navigation menu using HTML?
42. How do you add subtitles to a video using HTML?
43. What is the purpose of the action and method attributes in a form?
44. How do you create a dropdown list in a form?
45. What is the purpose of the <dl>, <dt>, and <dd> tags?
46. What is the <main> tag used for?
47. What is the <base> tag used for?
48. What is the purpose of the action attribute in a form?
49. What does the method attribute in a form do?
50. Explain the difference between GET and POST methods.
51. What is the use of the name attribute in input elements?
52. How can you create a dropdown list in an HTML form?

53. What are the different types of input elements in HTML forms?
54. How do you handle form validation in HTML?
55. What is the required attribute?
56. How do you create a radio button group?
57. What is the autocomplete attribute?
58. Explain the use of the placeholder attribute.
59. How can you create a file upload field?
60. What is the difference between reset and submit buttons?
61. How can you group related elements in a form?
62. What is the purpose of the label element?
63. How do you specify default values for form inputs?
64. What is form validation?
65. What are the HTML5 input types introduced?
66. Explain the novalidate attribute.
67. What is the purpose of the form attribute in input elements?
68. How do you create a multi-line text input?
69. What is the maxlength attribute?
70. How can you submit a form using JavaScript?
71. What is the purpose of the formaction attribute?
72. How do you create a form that supports file uploads?
73. What is the disabled attribute?
74. What is the pattern attribute?
75. How can you create a hidden input field?
76. What is the autofocus attribute?
77. What is the list attribute in input elements?
78. Explain the use of the min and max attributes.
79. What is the step attribute?
80. How can you create a search input field?
81. What is the accept attribute in file input?
82. How do you create a date input field?
83. What is the multiple attribute in file input?
84. How do you handle form data in JavaScript?

85. What is the enctype attribute?
86. Explain the for attribute in the label element.
87. How do you create a number input field?
88. What is the formmethod attribute?
89. How do you create a color input field?
90. What is the inputmode attribute?
91. How do you create a time input field?
92. Explain how to create a range slider.
93. What is the spellcheck attribute?
94. How do you create an email input field with validation?
95. How do you create an unordered list in HTML?
96. How do you create a telephone link in HTML?
97. How do you create an email link in HTML?
98. How do you link to an internal section within the same page?
99. What is the purpose of the action attribute in the <form> tag?
100. What is the difference between block-level and inline elements?
101. Use semantic HTML elements to create a simple webpage layout with a header, navigation, main content, and a footer.
102. Create a HTML form with client-side validation for required fields.
103. Create a form for TechInnovate's newsletter subscription with input fields for name, email, and a submit button.
104. Create a responsive webpage layout for Lakme using HTML and CSS (no frameworks).

CSS

(Total no. of Questions 102)

1. What is CSS?
2. How do you include CSS in an HTML document?
3. What are the different types of CSS?
4. What is the box model in CSS?
5. How do you center a block element horizontally?
6. What is the difference between padding and margin?
7. What is a CSS selector?
8. What are the different types of selectors in CSS?
9. What is the difference between class and ID selectors?
10. How do you apply multiple classes to an element in HTML?
11. What is specificity in CSS?
12. What is the !important declaration in CSS?
13. What are pseudo-classes in CSS?
14. What are pseudo-elements in CSS?
15. How do you create a CSS grid layout?
16. What is the flexbox layout in CSS?
17. How do you make a responsive design using CSS?
18. What is a CSS preprocessor?
19. What is a CSS framework?
20. What is the difference between em and rem units in CSS?
21. What are the different position values in CSS?
22. What is the z-index property in CSS?
23. How do you create a fixed header in CSS?
24. What is the float property used for?
25. How do you clear floated elements in CSS?
26. What is the display property and its possible values?
27. How do you create a two-column layout using CSS?
28. What is the overflow property in CSS?
29. How do you make an element invisible but still take up space in the layout?

30. What is the clip-path property in CSS?
31. How do you create a simple CSS transition?
32. What is the transform property in CSS?
33. How do you create a CSS animation?
34. What is the difference between transition and animation in CSS?
35. What is the animation-delay property?
36. What are CSS keyframes?
37. How do you create a bounce effect using CSS?
38. What is the @keyframes rule?
39. What is the transition-timing-function property?
40. How do you create a fade-in effect using CSS?
41. What is responsive web design?
42. What is mobile-first design in CSS?
43. What are CSS variables?
44. How do you use media queries in CSS?
45. What is the calc() function in CSS?
46. What are vendor prefixes in CSS?
47. What is BEM in CSS?
48. What are CSS frameworks and name a few popular ones?
49. What is the difference between inline, inline-block, and block elements?
50. What is the box-sizing property in CSS?
51. How can you optimize CSS for better performance?
52. What are CSS sprites?
53. How do you minify CSS?
54. What is critical CSS?
55. How do you use font-face in CSS?
56. What is the will-change property in CSS?
57. How can you defer loading of non-critical CSS?
58. What is the content-visibility property?
59. How do you improve rendering performance with CSS?
60. What are CSS modules?
61. What is CSS Grid?

62. How do you create a grid container in CSS?
63. What is the grid-template-areas property?
64. How do you use grid-template-columns and grid-template-rows?
65. What is the flex property in CSS?
66. How do you create a flex container in CSS?
67. What is the flex-direction property?
68. What is the justify-content property?
69. What is the align-items property?
70. What is the align-self property?
71. What is the calc() function in CSS?
72. What is the var() function in CSS?
73. What is the clamp() function in CSS?
74. What is the difference between px, em, rem, %, and vh/vw units?
75. .How do you use nth-child selector in CSS?
76. What is the nth-of-type selector?
77. What is the not() selector in CSS?
78. How do you use attr() in CSS?
79. What are viewport units in CSS?
80. How do you use custom properties (CSS variables) in CSS?
81. What are shorthand properties in CSS?
82. How do you use the margin shorthand property?
83. How do you use the padding shorthand property?
84. How do you use the border shorthand property?
85. How do you use the background shorthand property?
86. How do you use the font shorthand property?
87. How do you use the flex shorthand property?
88. How do you use the list-style shorthand property?
89. How do you use the transition shorthand property?
90. How do you use the animation shorthand property?
91. How do you debug CSS issues?
92. What are CSS resets?
93. How do you use the outline property for debugging?

94. What is the difference between outline and border?
95. How do you use the :hover pseudo-class for testing?
96. What are CSS custom properties (variables) and how do you debug them?
97. How do you test for responsive design?
98. What are CSS grid inspector tools?
99. How do you handle browser-specific CSS issues?
100. How do you validate your CSS?
101. Write CSS rules to set the font family to Arial, color to dark blue, and font size to 16px for all paragraphs.
102. Create a class named .highlight to change the background color to yellow and font weight to bold, and apply it to some text.